Modelling and Analysis

1D Project – Written Answers

3. (a)

The line generated using the linear regression function has equation $y = 73.334x + 10.777$, where y the vertical axis represents m_r, and x the horizontal axis represents the corresponding r^2

Mapping this to the equation $m_r = Ar^2 + B$, we get:
$A = 73.334$

$B = 10.777$

4. (a)

By using excel functions, we were able to calculate the averages of the various values.

Upright Stationary	$\overline{ax} = 0.92574\ m/s^2$ $\overline{ay} = 0.60313\ m/s^2$
Tilted Stationary	$\overline{az} = -0.07033\ m/s^2$

4. (b)

The following values were used to calculate a_r :

$\overline{ax} = 0.92574\ m/s^2$

$\overline{ay} = 0.60313\ m/s^2$

$\overline{az} = -0.07033 m/s^2$

$\overline{ax}$ and $\overline{ay}$ are referenced from the “Upright Stationary”. $\overline{az}$ is referenced from the “Tilted Stationary” sheet.

a_r is calculated using the following formula:

$$a_r = \sqrt{(ax - \overline{ax})^2 + (ay - \overline{ay})^2 + (az - \overline{az})^2}$$

4. (c)

$\overline{a_r} = 9.99560$

4. (d)

a is calculated using the formula below:

$$a = -(\sqrt{(ax - \overline{ax})^2 + (ay - \overline{ay})^2 + (az - \overline{az})^2} - \overline{a_r})$$

It is applied by creating a column adjacent to the az column and applying the formula for each row of data.

4. (e)

To properly set bounds for each of the graphs and sheets, the time where the cargo is dropped and the time where the cargo impacts the ground.

These are indicated on each sheet by the cells highlighted in colour.

- Time where cargo is dropped in Yellow
- Time where cargo impacts ground in Red

Below is a table containing all the timings identified:

Sheet Name	Time Cargo is Dropped (μs)	Time Cargo Impacts Ground (μs)
40 cm 1st	3788004	6386004
40 cm 2nd	4244004	7024000
45 cm 1st	3228000	6200000
45 cm 2nd	4678000	7402000
50 cm 1st	2812000	6055996
50 cm 2nd	1898004	4988004
55 cm 1st	2180000	5548000
55 cm 2nd	4990000	8248000
60 cm 1st	5076000	8624008
60 cm 2nd	4440000	8131996
65 cm 1st	2574000	6444000
65 cm 2nd	5348000	8794000
70 cm 1st	3664000	7356000
70 cm 2nd	6758000	10260004
75 cm 1st	4356000	8124008
75 cm 2nd	6001992	9525996
80 cm 1st	6430004	11018000
80 cm 2nd	6596000	10549996

4. (f)

By assuming that time at time which cargo is dropped, $v = 0$ we can calculate the velocity at each row of data using the time passed and previously calculated velocity.

An additional column of data where between the Time Cargo is Dropped and Time Cargo Impacts Ground is added, where the velocity is calculated and show in the spreadsheet.

As the time is given in microseconds while the acceleration is given in seconds, the functions used to calculate the velocity must also account for the unit of measurement for time passed.

4. (g)

Terminal velocity **(v_T)** is simply the velocity calculated at the point of impact.

Sheet Name	$\boldsymbol{v_T\ (m/s)}$
40 cm 1st	1.92466
40 cm 2nd	1.67574
45 cm 1st	1.61538
45 cm 2nd	0.93083
50 cm 1st	1.84862
50 cm 2nd	1.57646
55 cm 1st	1.12788
55 cm 2nd	1.53476
60 cm 1st	1.35432
60 cm 2nd	1.55735
65 cm 1st	1.07263
65 cm 2nd	1.06190
70 cm 1st	1.41066
70 cm 2nd	0.49285
75 cm 1st	-0.12401
75 cm 2nd	0.92404
80 cm 1st	0.84740
80 cm 2nd	0.92719

4. (h)

v_{Ti} for each r_i is calculated by calculating the average of the two values of $\boldsymbol{v_T}$ obtained for each parachute radius (XX cm 1st and XX cm 2nd).

$$Example\ Using\ 40cm\ Parachute\!:\ \ r_{0.4m} = \frac{1.92466 + 1.67574}{2}$$

$\boldsymbol{r_i}$	$\boldsymbol{v_{Ti}}$
0.40m	1.8002
0.45m	1.273105
0.50m	1.71254
0.55m	1.33132
0.60m	1.455835
0.65m	1.067265
0.70m	0.951755
0.75m	0.92404
0.80m	0.887295

Note: Value of $\boldsymbol{v_T}$ for 75 cm 1st is omitted in calculation as it is negative, suggesting an error.

4. (i)

To differentiate function $e(C)$, we decided to simplify the known values (In this case A, B, r_i, v_{Ti}, and any other constant) as they are functionally identical to constants next to the variable C.

$$\boldsymbol{e(C) := \sum_{i=1}^{9}(v_{Ti} - \sqrt{\frac{(Ar_i^2 + B)g}{C}}\frac{1}{r_i})^2}$$

$$\boldsymbol{X = v_{Ti}}$$

$$\boldsymbol{\sqrt{\frac{(Ar_i^2 + B)g}{C}}\frac{1}{r_i} = \sqrt{(Ar_i^2 + B)g}\sqrt{C^{-1}}\frac{1}{r_i}}$$
$$\boldsymbol{= \frac{1}{r_i}\sqrt{(Ar_i^2 + B)g}\, C^{-\frac{1}{2}}}$$

$$\boldsymbol{Y = \frac{1}{r_i}\sqrt{(Ar_i^2 + B)g}}$$

In this case, the constants X and Y are created. This is done by taking the components of the function which are not variable C. Substituting these constants into the function we get the following:

$$e(C) = \sum_{i=1}^{9}(X - YC^{-\frac{1}{2}})^2$$

This simplified equation allows up to find the first and second derivatives without dealing with all the different variables.

$$\begin{aligned}
e'(C) &= \frac{d}{dC}\sum_{i=1}^{9}\left(X - YC^{-\frac{1}{2}}\right)^2 \\
&= \sum_{i=1}^{9}\frac{d}{dC}\left(X - YC^{-\frac{1}{2}}\right)^2 \\
&= \sum_{i=1}^{9}2\left(X - YC^{-\frac{1}{2}}\right)\left(\frac{YC^{-\frac{3}{2}}}{2}\right) \\
&= \sum_{i=1}^{9}\left(X - YC^{-\frac{1}{2}}\right)\left(YC^{-\frac{3}{2}}\right) \\
&= \sum_{i=1}^{9}(XYC^{-\frac{3}{2}} - Y^2C^{-2})
\end{aligned}$$

$$e''(C) = \frac{d}{dC}\sum_{i=1}^{9}(XYC^{-\frac{3}{2}} - Y^2C^{-2})$$

$$= \sum_{i=1}^{9} \frac{d}{dC}(XYC^{-\frac{3}{2}} - Y^2C^{-2})$$

$$= \sum_{i=1}^{9} (-\frac{3}{2}XYC^{-\frac{5}{2}} + 2Y^2C^{-3})$$

Now that we have the first and second derivatives of $e(C)$, we use the first derivative to identify the critical point, $C*$. Using $C*$ found, we can perform the second derivative test to prove that the critical point is the global minimiser.

Since gradient is 0 at critical point, $r(C*) = 0$

$$e(C^*) \Rightarrow \sum_{i=1}^{9} (XY(C^*)^{-\frac{3}{2}} - Y^2(C^*)^{-2}) = 0$$

$$e'(C^*) \Rightarrow \sum_{i=1}^{9} (XY(C^*)^{-\frac{3}{2}}) = \sum_{i=1}^{9} (Y^2(C^*)^{-2})$$

$$(C^*)^{-\frac{3}{2}} \sum_{i=1}^{9} (XY) = (C^*)^{-2} \sum_{i=1}^{9} (Y^2)$$

$$(C^*)^{\frac{1}{2}} = \frac{\sum_{i=1}^{9}(Y^2)}{\sum_{i=1}^{9}(XY)}$$

$$\sqrt{C^*} = \frac{\sum_{i=1}^{9}(Y)}{\sum_{i=1}^{9}(X)}$$

$$C^* = \left(\frac{\sum_{i=1}^{9}(Y)}{\sum_{i=1}^{9}(X)}\right)^2$$

$$= \left(\frac{\sum_{i=1}^{9}\left(\boldsymbol{\frac{1}{r_i}\sqrt{(Ar_i^2 + B)g}}\right)}{\sum_{i=1}^{9}(v_{Ti})}\right)^2$$

If we were to calculate the summations after substituting the values obtained in all previous steps, we would be able to find the value C^* at the critical point.

To prove that the critical point is a global minimiser, we substitute C^* into the second derivative.

$$e''(C^*) = \sum_{i=1}^{9} (-\frac{3}{2}XY\left(\left(\frac{Y}{X}\right)^2\right)^{-\frac{5}{2}} + 2Y^2\left(\left(\frac{Y}{X}\right)^2\right)^{-3})$$

$$= \sum_{i=1}^{9} \left(-\frac{3}{2} XY \left(\frac{Y}{X}\right)^{-5} + 2Y^2 \left(\frac{Y}{X}\right)^{-6} \right)$$

$$= \sum_{i=1}^{9} \left(-\frac{3}{2} XY \left(\frac{X}{Y}\right)^{5} + 2Y^2 \left(\frac{X}{Y}\right)^{6} \right)$$

$$= \sum_{i=1}^{9} \left(-\frac{3}{2} \left(\frac{X^6}{Y^4}\right) + 2 \left(\frac{X^6}{Y^4}\right) \right)$$

$$= \sum_{i=1}^{9} \left(\frac{1}{2} \left(\frac{X^6}{Y^4}\right) \right)$$

$$= \sum_{i=1}^{9} \left(\frac{X^6}{2Y^4} \right)$$

The above is the result of substituting C^* into second derivative. The value can be proven to be always positive using the following:

$\boldsymbol{X = v_{Ti}}$

X solely comprises of v_{Ti}, which is always a positive value. This is because the calculation of v_{Ti} ruled that all negative values are deemed as errors and are not to be accounted for in calculations.

$\boldsymbol{Y = \frac{1}{r_i}\sqrt{(Ar_i^2 + B)g}}$

Remember in question 3(a) where $m_r = Ar^2 + B$, which is always positive.

$\boldsymbol{r_i}$ is also always a positive value as it is a measurement of the radius of the parachute.

Therefore, for all cases the value of $Y > 0$.

With the above understanding. It can be concluded that:

$$\frac{\sum_{i=1}^{9}(X^6)}{\sum_{i=1}^{9}(Y^4)} > 0$$

$$e''(C^*) = \sum_{i=1}^{9} \left(\frac{X^6}{2Y^4} \right) > 0$$

By second derivative test, critical point C^* is the global minimiser.